

Pilot Study: Fat Pad Restoration With Adipose Allograft in Metatarsalgia Patients Shows Structural, Functional Improvement

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Background: Metatarsalgia is a painful and debilitating condition that is often linked to plantar fat pad atrophy, particularly in active individuals and those with structural foot deformities. Current treatment options range from conservative measures—such as orthotics and padding, which often provide only temporary relief—to more invasive surgical corrections. Fat pad restoration has emerged as a promising nonsurgical solution. Human adipose tissue allograft (hATA) is a novel therapeutic option for reconstruction or supplementation of adipose defects or damage.

Methods: This retrospective case series included 9 patients (10 feet) with metatarsalgia secondary to plantar fat pad atrophy who underwent in-office implantation with 1.5 mL of hATA between August and November 2024. All procedures were performed using an 18G blunt-tip cannula, with offloading protocols postinjection. Outcomes were assessed at 12 weeks using ultrasound-based measurements of fat pad thickness and Foot and Ankle Disability Index scores.

Results: At 12 weeks, 9 of 10 treated feet showed increased fat pad thickness (mean gain 0.21 cm, 65% improvement). Foot and Ankle Disability Index scores improved in 8 of 9 patients, with a mean increase from 66.0 to 84.2 (18.1 points); 56% of patients exceeded the minimum clinically important difference of 10 points. No statistically significant correlation was observed between tissue thickness gain and functional improvement (Pearson $r = -0.38$).

Conclusions: Use of hATA seems to be a safe, nonsurgical option with potential to restore cushioning and reduce symptoms in patients with metatarsalgia. (*Plast Reconstr Surg Glob Open* 2025;13:e7336; doi: [10.1097/GOX.00000000000007336](https://doi.org/10.1097/GOX.00000000000007336); Published online 3 December 2025.)

INTRODUCTION

The fat pad of the foot is the natural layer of subcutaneous cushioning found on the bottom of the foot, thickest beneath the ball and heel. Plantar fat pad atrophy occurs when the subcutaneous padding is damaged, displaced, or degenerated, significantly reducing cushioning and shock absorption.^{1,2} This tissue loss can make weight-bearing activities; wearing a variety of footwear; and maintaining a healthy, active lifestyle increasingly difficult. In more severe cases, fat pad atrophy can become debilitating, making even simple tasks

such as standing barefoot in the shower painful and challenging.³

Fat pad atrophy can present across a broad range of patients, affecting both younger and older, otherwise healthy individuals and those with underlying systemic conditions. In younger patients, it is often posttraumatic and may be associated with plantar plate tears, bunions, Tailor's bunions, crush injuries, or neuromas. Iatrogenic causes include cortisone-induced lipoatrophy and surgical scarring. In older individuals, contributing factors include aging, autoimmune diseases, diabetic neuropathy, progressive cavus foot deformities, and certain medications.^{3–8} Approximately 30% of patients aged 60 years or older are affected by forefoot fat pad atrophy.⁹ A common clinical presentation is heel pain or metatarsalgia, particularly in

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Limitations regarding long-term follow-up inherently exist in this article type.

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older adults, physically active individuals, and those with structural foot deformities.⁸

Diagnosis is typically based on clinical history, physical examination, and/or imaging. Patients often report forefoot or heel pain that worsens with barefoot walking or weight-bearing on hard surfaces. On examination, increased prominence of bony landmarks and tenderness to palpation may be noted, and diagnosis is supported by documented thinning of the plantar soft tissue on ultrasound or magnetic resonance imaging (MRI).^{10,11}

Once diagnosed, treatment decisions are guided by the severity of symptoms, underlying pathology, and patient-specific factors. Conservative measures such as padding, footwear modifications, and shock-absorbing shoes may provide symptom relief, but often require daily maintenance and lifestyle adjustments.¹² On the other end of the spectrum, surgical intervention may be required to address contributing bone, tendon, or nerve pathology when conservative therapies fail.¹³

Fat pad restoration serves as an intermediary solution between conservative management and surgical correction and has gained increasing acceptance in clinical practice as research and clinical data have demonstrated its effectiveness not only in reducing pain during ambulation but also in improving overall quality of life.^{14–23} In-office injections—typically involving the placement of a cushioning material into areas of fat pad loss—represent a low-risk, relatively painless, and appealing option for patients seeking more durable relief without the burden of ongoing daily modifications.

Various options of these in-office therapies for fat pad restoration have been explored, including the use of silicone, dermal fillers, autologous fat grafting, and adipose allograft matrices (AAMs).^{14–23} More recently, a human adipose tissue allograft (hATA), which retains the structure and the cushioning function of native adipose tissue, has been developed as a novel alternative.²⁴ Here, we describe the use of hATA in patients with metatarsalgia secondary to fat pad atrophy.

METHODS

hATA Product Description

The hATA used (Liposana, Britecyte, Inc., Frederick, MD) is a commercially available human tissue product that preserves the native architecture of adipose tissue. It is regulated by the US Food and Drug Administration (FDA) as a human cells, tissues, and cellular and tissue-based product under Section 361 of the Public Health Service Act and Title 21 of the Code of Federal Regulations Part 1271.²⁵

Manufacturing of hATA is done in collaboration with LifeLink Tissue Bank; LifeLink is accredited by the American Association of Tissue Banks, registered with the FDA and Health Canada; licensed by the states of Florida, California, Maryland, and New York; and registered in Delaware and Oregon. The hATA is processed from donated subcutaneous abdominal adipose tissue obtained from cadaveric donors. Donor screening, tissue recovery, processing, and distribution are performed in compliance

Takeaways

Question: Can supplementation of a degenerated plantar fat pad with human adipose tissue allograft (hATA) in patients with metatarsalgia improve fat pad tissue thickness and functionality?

Findings: In this retrospective pilot study of 9 patients (10 feet), hATA implantation resulted in increased plantar thickness in 90% of treated feet and improved Foot and Ankle Disability Index scores in 89% of patients at 12 weeks. No adverse events were reported. Notably, structural fat pad tissue gains did not correlate with functional improvement.

Meaning: The use of hATA may offer a safe, minimally invasive option for fat pad restoration, with rapid improvements in pain and function in patients with metatarsalgia.

with applicable FDA, American Association of Tissue Banks, state, local, and other applicable regulations.²⁵ The hATA is supplied in a 1.5-mL volume within a sterile cryogenic vial featuring a screw cap, packaged within dual chevron-type peel pouches. The shelf life of hATA is 5 years if stored at -40°C or colder. Storage between 0°C and 10°C allows for 12 months of shelf life.²⁵ The hATA used in this study was administered in an office-based setting as a cash-pay procedure.

Study Design and Patient Population

Patients diagnosed with metatarsalgia secondary to fat pad atrophy who received hATA implantation between August and November 2024 were included in this retrospective analysis. All patients presented to the investigator's clinic with painful forefoot pathology and were evaluated using medical history, physical examination (including palpation), and ultrasound assessment. In select cases, MRI was also performed to assess for ligament tears, stress fractures, or tendonitis. None of the patients had any underlying acute or active infections, cancerous lesions, or bone fractures. Following evaluation, patients were counseled on various treatment options, including offloading pads, orthotic insoles, oral anti-inflammatory medications, fat pad restoration, and surgical intervention. Informed consent was obtained from each patient before implantation. De-identified data were collected from medical charts in accordance with the Health Insurance Portability and Accountability Act of 1996. The study adhered to the ethical principles outlined in the Declaration of Helsinki, and institutional review board approval was waived due to the retrospective nature of the analysis.

hATA Implantation Technique

All hATA implantations were performed in an office setting. A local block was attained using 1% plain lidocaine. The skin was prepared with Betadine or Hibiclens solution, and a sterile no. 11 blade was used to create a small stab incision. An 18G blunt-tip microcannula was then used to deliver the adipose tissue in a fan-like pattern, depositing microdroplets along the desired path within the subcutaneous layer. In all cases, the graft

was targeted to the central plantar forefoot region, specifically beneath the metatarsal heads, as identified on preprocedure ultrasound. Care was taken to evenly distribute the graft across the affected weight-bearing surface. The incision site was closed with Steri-Strips, and patients were provided with offloading pads following the procedure.

Postimplantation Management and Clinical Outcomes

Patients were instructed to offload and to limit weight-bearing activity for 2 weeks following the procedure. This included wearing cushioned offloading footwear, avoiding barefoot walking, strenuous exercise, or prolonged standing, and refraining from the use of unsupportive footwear. Ambulation for activities of daily living was permitted. Follow-up evaluations were conducted at 2 and 12 weeks postimplantation. Outcome measures included comparison of pre- and 12-week postimplant Foot and Ankle Disability Index (FADI) scores, with a minimal clinically important difference (MCID) defined as ± 10 points. Plantar fat pad thickness and tissue quality were also assessed using ultrasound, with measurements obtained from the dermal surface to the cortical boundary of the underlying bone, including any intervening tendon structures.

RESULTS

A total of 9 patients (10 feet) received hATA implants between August and November 2024. The cohort included 6 women (67%) and 3 men (33%), with a mean age of 65.8 years (range: 51–76 y). Common comorbidities included diabetes mellitus ($n = 3$) and prior foot deformities or surgical interventions ($n = 7$). All patients received 1.5 mL of hATA per site, with implant locations targeting submetatarsal fat pads (submet 1–5). Four patients (5 feet) had pain in the first submetatarsal, 3 patients (3 feet) in the second submetatarsal, and 2 patients (2 feet) received hATA in the fifth submetatarsal. [Table 1](#) provides a summary of patient demographics and implant details.

Ultrasound measurements demonstrated an average baseline fat pad thickness of 0.56 cm (range: 0.24–1.18 cm), increasing to an average of 0.77 cm at 12 weeks (range: 0.41–1.28 cm). The mean absolute plantar fat pad thickness gain was 0.21 cm, corresponding to an average tissue thickness percent improvement of 65%. Nine of 10 implanted feet (90%) showed increased thickness; 1 foot demonstrated a decrease (–15%) ([Table 2](#)).

The mean preimplant FADI score was 66.0 (range: 31–98), increasing to 84.1 at 12 weeks postimplant (range: 23–104), reflecting a mean improvement of 18.1 points. Eight of 9 (89%) patients showed improvement

Table 1. Patient Demographics, Clinical History, and Implant Details

Case	Sex	Age, y	Significant Medical History	Prior Treatments	Implant Location (1.5 mL)
1	M	75	None	Surgical planning submet 5 y ago	R submet 5
2	F	62	Noninsulin-dependent diabetes mellitus, obesity, bunion, hammertoes, arthritis of the great toe joint	Bunion surgery 23 y ago, debridement of callus	L submet 5
3	M	68	Venous insufficiency, cavus foot, submet 2 plantar plate tear	Change in shoes	L submet 2/3
4	F	73	Hallux limitus with elevatus	Orthotics, fat pad restoration with dermal filler 2.5 y ago	R submet 2/3
5	F	55	None	Bunion surgery, cortisone injection, metatarsal pad, walking boot, prefabricated orthotics	R submet 2
6	F	74	Insulin-dependent diabetes mellitus, kidney disease, thyroid disease, bunion, hammertoes	Acetaminophen, offloading pads	R submet 1
7	F	76	Sesamoid issues	Orthotics, change in shoes	R submet 1
8	M	51	Sesamoid fracture 7 y ago, continued sesamoid pain, plantar plate tear R submet 2	Orthotics, change in shoes, oral steroid, immobilization	R submet 1
9	F	58	Rheumatoid arthritis, bunion, arthritis of the great toe joint	Cortisone injection into joint, dermal filler in ball of the foot, AAM	R submet 1/L submet 1

F, female; L, left; M, male; R, right.

Table 2. Imaging and Functional Outcomes

Case	Baseline Thickness, cm	12-wk Thickness, cm	Thickness Difference, cm	% Thickness Improvement	Baseline FADI Score	12-wk FADI Score	FADI Difference
1	0.25	0.41	0.16	64	68	100	32
2	0.27	0.61	0.34	126	98	100	2
3	1.18	1.28	0.10	8	68	72	4
4	0.90	1.12	0.22	24	50	104	54
5	1.03	0.88	–0.15	–15	61	91	30
6	0.29	0.41	0.12	41	31	23	–8
7	0.42	0.55	0.13	31	85	102	17
8	0.49	0.66	0.17	35	72	98	26

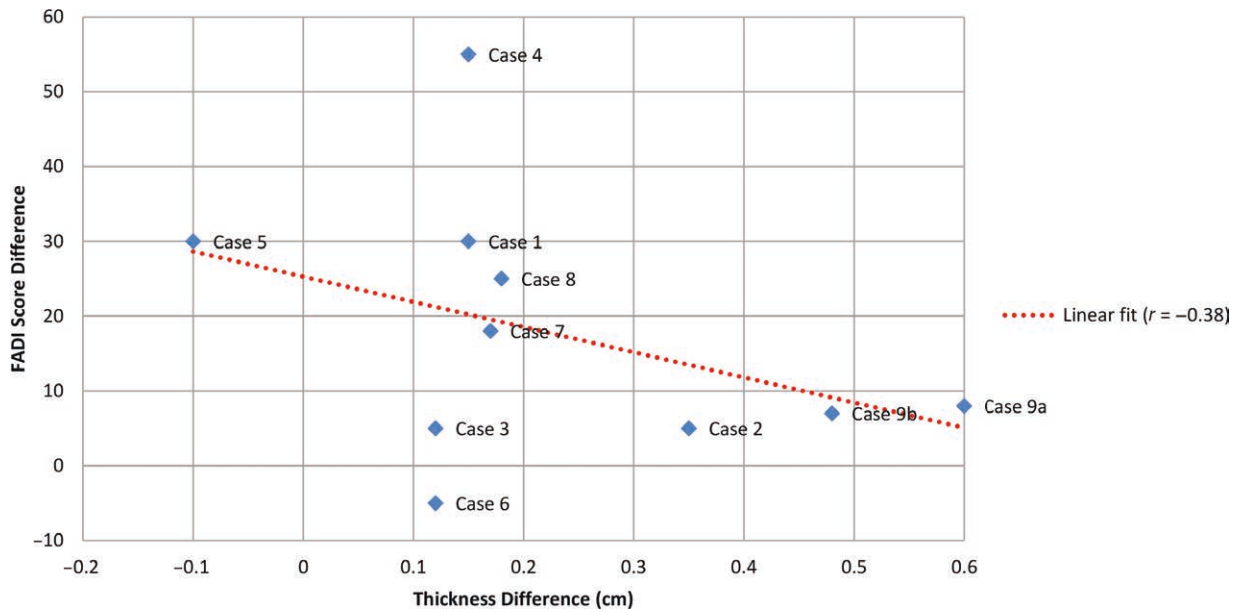


Fig. 1. Correlation between plantar fat pad thickness improvement and FADI score change at 12 weeks postimplant. Each dot represents an individual foot. Although some cases showed concurrent anatomical and functional gains, the overall correlation was weak.

in FADI score at 12 weeks, and 5 (55.6%) patients exceeded the MCID threshold of ± 10 points (Table 2). One patient experienced an 8-point decrease in FADI, which did not meet the MCID. Another patient had a high baseline score of 98, leaving limited room for measurable improvement within the scale's upper range. Despite the high functional score for this patient, fat pad atrophy was confirmed on physical examination and ultrasound, and the patient reported experiencing pain anecdotally. Notably, there was no statistically significant correlation between the change in tissue thickness and FADI score improvement (Pearson $r = -0.38$, $P = 0.39$), suggesting that although hATA implantation may provide structural cushioning, perceived functional benefit may be influenced by other factors (Fig. 1). No product-related adverse events were reported during the study period. A representative case is described as follows.

Case 8

A 51-year-old man with a history of right sesamoid fracture, chronic sesamoid pain, and a second plantar plate tear presented with longstanding discomfort localized beneath the right first submetatarsal head. Prior treatments included custom orthotics, oral corticosteroids, and periods of immobilization, all of which provided only temporary relief.

The patient received a 1.5-mL hATA implant at the painful submetatarsal 1 site, followed by offloading with adhesive plantar padding (Fig. 2). Baseline ultrasound demonstrated a preimplant plantar fat pad thickness of 0.49 cm. At 12 weeks postimplantation, thickness increased to 0.66 cm, representing a 0.17 cm gain and a 35% relative thickness improvement (Fig. 3). Postprocedural imaging also showed increased internal

echogenicity compared with baseline, suggestive of enhanced tissue density and remodeling at the implantation site (Fig. 3).

Functionally, the patient's FADI score improved from 72 to 98, reflecting a 26-point increase and a clinically meaningful improvement in pain and activity tolerance. Notably, the patient reported complete resolution of previously recurrent flare-ups and pain episodes during the 12-week posttreatment period—symptoms that had been persistent before implantation. When asked to estimate overall satisfaction at 12 weeks, the patient reported being very satisfied.

DISCUSSION

Fat pad restoration has emerged as a nonsurgical treatment option for patients experiencing pain due to plantar fat pad atrophy, particularly in the context of metatarsalgia. Although clinical interest in fat pad restoration is increasing, effective and practical treatment options remain limited. Several modalities have been explored in the literature, including silicone injections, dermal fillers, autologous fat grafting, and adipose-derived matrices.^{14–23}

One of the earliest studies in this space was a randomized, double-blind controlled trial by van Schie et al,¹⁵ which evaluated liquid silicone injections in diabetic neuropathic feet to reduce ulceration risk. Twenty-eight patients received multiple silicone injections into the plantar surface, resulting in a significant increase in soft tissue thickness and a corresponding decrease in plantar pressure that was sustained for up to 12 months.¹⁵ However, as the study population was neuropathic, pain and function were not assessed.



Fig. 2. Plantar forefoot of a patient with metatarsalgia and fat pad atrophy. A, Preimplantation image showing the plantar surface before subcutaneous injection of hATA. B, Immediate postimplantation image, with adhesive plantar padding applied for offloading and protection of the implanted region.

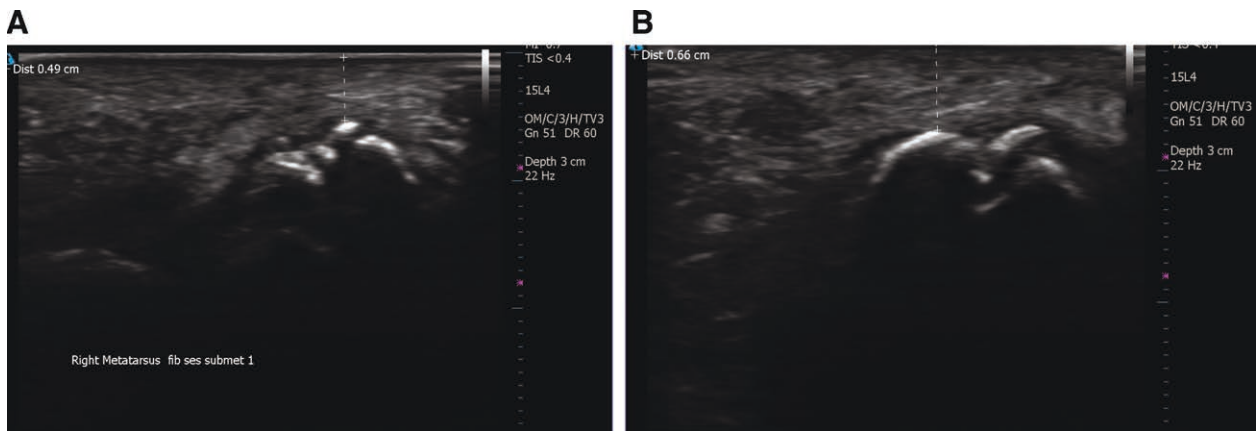


Fig. 3. Pre- and postimplant ultrasound images demonstrating changes in plantar soft tissue echogenicity beneath the right first submetatarsal head. A, Baseline: the fat pad seems relatively hypoechoic with limited internal architecture, consistent with atrophic, low-density tissue. B, 12 weeks postimplantation: increased echogenicity and internal echoes are observed, suggesting early tissue remodeling and improved structural organization following hATA implantation.

Subsequent studies explored the use of dermal fillers. In 2018, Fountente et al¹⁶ evaluated a hyaluronic acid filler in 15 patients with high-heel-associated metatarsalgia and reported improved pain tolerance and reduced overall pain at 6 months. A 2022 study using dermal filler (Juvederm Voluma XC, Allergan Aesthetics, Irvine, CA) in 30 patients demonstrated significant improvement in FADI scores over a 54-week period.²¹ Despite these findings, neither silicone nor dermal fillers are FDA-cleared for use in the foot and both lack the structural and biomechanical properties of native adipose, potentially limiting their utility.

Autologous fat grafting is another therapeutic option for fat pad restoration that has been previously described in the literature. In a 2017 pilot study of 8 patients with heel pain following adult flatfoot surgery, subcutaneous fat grafting led to a significant reduction in pain scores, with the mean decreasing from 8.13 at baseline to 2.41 at 6 months postprocedure ($P < 0.001$).²⁶ In another pilot study, Minter et al²² compared 2 cohorts: 1 underwent fat grafting with a 2-year follow-up, whereas the other received conservative management for 12 months before undergoing the same grafting procedure. Both groups reported immediate pain relief following grafting. However, only

the early-intervention group experienced sustained improvements at 12, 18, and 24 months.²² Although outcomes were promising, autologous grafting requires fat harvesting and processing, which introduces the potential for donor site morbidity and carries significant variability in graft retention over time.^{27–30}

AAM has been explored as a less invasive alternative to autologous grafting. In a small 16-patient study (21 feet), including 13 cases of submetatarsal pain, AAM was associated with a 72.9% average improvement in clinical outcomes after 7.7 months.²³ AAM is decellularized and delipidized, lacking adipocytes—the primary contributors to native fat pad cushioning. Its effects rely on host remodeling, and full integration may take several months.³¹

More recently, advances in preservation technology have led to the development of an hATA, which retains both the structural matrix and adipocytes of native adipose tissue.²⁴ Use of hATA offers the potential for both volumetric restoration and biomechanical cushioning. In a prior pilot study, hATA was used in 12 diabetic neuropathic patients with preulcerative plantar lesions. Over an average 6.4-month follow-up (up to 10 mo), no reulcerations were reported, and the affected areas demonstrated visible clinical improvement.²⁴ Although pain was not formally assessed, the study supported the safety and potential benefits of hATA for volume restoration of the fat pad.

The present case series is the first to evaluate hATA in patients with symptomatic metatarsalgia, with a focus on patient-reported pain, functional improvement, and changes in plantar tissue thickness. Most treated feet showed increased plantar thickness at 12 weeks (mean gain: 0.21 cm), and nearly all patients reported functional improvement as measured by FADI scores. Over half of the patients (56%) exceeded the MCID threshold of 10 points. These results support hATA's potential as a minimally invasive, office-based intervention for patients with symptomatic fat pad atrophy.

Normal plantar fat pad thickness in the foot is generally reported to fall within the 1–2 cm range in healthy adults, though this may vary based on regional anatomical differences.^{32–35} In our study, 8 of the 10 treated sites had baseline fat pad thickness values below this range, consistent with fat pad atrophy. However, 2 cases—case 3 (1.18 cm) and case 5 (1.03 cm)—fell within the lower end of the normal range at baseline. Interestingly, both patients reported baseline FADI scores in the 60s, indicating significant functional impairment despite having “normal” fat pad thickness. Moreover, these cases showed minimal improvement in thickness at 12 weeks (case 3: +0.10 cm; case 5: –0.15 cm), suggesting that their initial ultrasound measurements may have been confounded by transient inflammation or soft tissue swelling. These findings underscore a broader observation: patient-reported pain and functional limitation may not always align with fat pad thickness.

This disconnect was further reflected in the overall lack of statistically significant correlation between anatomical restoration and patient-reported outcomes in our cohort. Although 9 of 10 treated feet showed increased plantar thickness, the Pearson correlation coefficient

between thickness gain and FADI score improvement was –0.38 (Fig. 1), indicating that structural gains did not reliably predict clinical benefit. For example, case 4 demonstrated a 54-point increase in FADI score despite only a moderate increase in tissue thickness, highlighting the subjective and multifactorial nature of pain and function. Although there is no correlation between fat pad thickness and functional outcomes, fat pad measurements in this study provide valuable information regarding changes in fat pad thickness and quality after hATA implantation.

This observation aligns with findings from Ruane et al,³⁶ who conducted volumetric MRI analysis on 17 patients undergoing autologous fat grafting. That study showed an initial increase in plantar thickness following grafting, followed by a return toward baseline levels over 2 to 6 months. Notably, this decrease in measurable thickness did not correspond with a loss of clinical benefit; patients continued to report improvements in pain and function.³⁶ Minter et al²² reported similar findings in their 2-year fat grafting study, where tissue thickness decreased toward baseline levels over 2 to 6 months, yet sustained pain relief was observed out to 24 months. Interestingly, the MRI analysis by Ruane et al³⁶ confirmed that although thickness declined, graft volume was retained, suggesting tissue redistribution rather than complete resorption.

In line with these reports, our findings suggest that implanted adipose tissue does not simply vanish but may shift, remodel, or redistribute pressure away from high-impact zones. However, it is also possible that a portion of the grafted fat may undergo atrophy or necrosis over time. As such, clinical outcomes in this setting are likely influenced by a combination of factors, including tissue viability, biomechanics, offloading compliance, pain modulation, and underlying comorbidities.

These results underscore the importance of prioritizing patient-reported outcomes and clinical findings—such as resolution of callus or hyperkeratosis—over imaging alone when assessing response. Ultrasound may be a helpful tool, but it should not be the sole determinant of treatment efficacy or the need for repeat implantation.

The limitations of this study include its small sample size, retrospective design, and a short follow-up window limited to 12 weeks. This limited duration may not fully capture the long-term durability, remodeling, or integration of the implanted tissue, especially when compared with autologous fat grafting studies that assess outcomes at 12 months or longer. The heterogeneity of patient comorbidities and prior interventions also introduces confounding factors. In addition, the 12-week endpoint may not capture the full remodeling and tissue integration process. Finally, FADI scores, although widely used, are still subject to interpatient variability and external influences. For example, 1 patient (case 2) reported a baseline FADI score of 98, despite presenting with clinically meaningful forefoot discomfort and ultrasound-confirmed fat pad atrophy. The decision to treat was based on symptom history, objective findings, and shared decision-making. This highlights a limitation of global functional scores such as the FADI, which may not adequately reflect localized symptoms such as mechanical pain

or activity-related pressure discomfort. Furthermore, patient-reported outcome measures are subject to psychological and behavioral influences—including pain tolerance, mood, and perception of function—which can affect score interpretation.

Despite these limitations, this case series provided early evidence that hATA may be a promising middle ground between conservative management and surgical reconstruction. By restoring cushioning and improving pain through a single in-office procedure, hATA offers a therapeutic option for patients with plantar fat pad atrophy and metatarsalgia who are seeking relief without invasive intervention. Notably, this cohort represents some of the first patients treated with hATA for this indication, and the primary objective was to evaluate early safety, feasibility, and short-term functional outcomes. These preliminary findings are hypothesis-generating and support the need for larger, prospective studies to validate long-term efficacy and durability.

CONCLUSIONS

This pilot study provided early evidence supporting the safety and clinical utility of hATA for fat pad restoration in patients with metatarsalgia secondary to plantar fat pad atrophy. The implantation was well tolerated, with no adverse events reported, and the majority of patients experienced meaningful improvements in pain and function following a single in-office procedure. Although functional outcomes did not consistently correlate with ultrasound-measured thickness gains, these findings highlight the multifactorial nature of symptom relief and the importance of patient-reported outcomes in evaluating treatment success. Future prospective studies with longer follow-up and larger sample sizes are warranted to validate these results.

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DISCLOSURES

Schoenhaus Gold serves as a consultant for Britecye, Inc., the company that markets hATA. Schoenhaus Gold is also a consultant for MTF Biologics. No ghostwriters were used in the writing of this article.

REFERENCES

1. Dalal S, Widgerow AD, Evans GR. The plantar fat pad and the diabetic foot—a review. *Int Wound J*. 2015;12:636–640.
2. Jahss MH, Michelson JD, Desai P, et al. Investigations into the fat pads of the sole of the foot: anatomy and histology. *Foot Ankle*. 1992;13:233–242.
3. Buldt AK, Forghany S, Landorf KB, et al. Foot posture is associated with plantar pressure during gait: a comparison of normal, planus and cavus feet. *Gait Posture*. 2018;62:235–240.
4. Fontanella CG, Nalesso F, Carniel EL, et al. Biomechanical behavior of plantar fat pad in healthy and degenerative foot conditions. *Med Biol Eng Comput*. 2016;54:653–661.
5. Murray HJ, Young MJ, Hollis S, et al. The association between callus formation, high pressures and neuropathy in diabetic foot ulceration. *Diabet Med*. 1996;13:979–982.
6. Kwan RL, Zheng YP, Cheing GL. The effect of aging on the biomechanical properties of plantar soft tissues. *Clin Biomech*. 2010;25:601–605.
7. Hsu TC, Wang CL, Tsai WC, et al. Comparison of the mechanical properties of the heel pad between young and elderly adults. *Arch Phys Med Rehabil*. 1998;79:1101–1104.
8. DiPreta JA. Metatarsalgia, lesser toe deformities, and associated disorders of the forefoot. *Med Clin North Am*. 2014;98:233–251.
9. Hannan MT, Menz HB, Jordan JM, et al. High heritability of hallux valgus and lesser toe deformities in adult men and women. *Arthritis Care Res (Hoboken)*. 2013;65:1515–1521.
10. Morrison T, Jones S, Causby RS, et al. Reliability of ultrasound in evaluating the plantar skin and fat pad of the foot in the setting of diabetes. *PLoS One*. 2021;16:e0257790.
11. Ricatti G, Veronese N, Gangai I, et al. Hoffa's fat pad thickness: a measurement method with sagittal MRI sequences. *Radiol Med*. 2021;126:886–893.
12. Choo YJ, Park CH, Chang MC. Rearfoot disorders and conservative treatment: a narrative review. *Ann Palliat Med*. 2020;9:3546–3552.
13. Luu CA, Larson E, Rankin TM, et al. Plantar fat grafting and tendon balancing for the diabetic foot ulcer in remission. *Plast Reconstr Surg Glob Open*. 2016;4:e810.
14. Balkin SW. Injectable silicone and the foot: a 41-year clinical and histologic history. *Dermatol Surg*. 2005;31:1555–1559; discussion 1560.
15. van Schie CH, Whalley A, Vileikyte L, et al. Efficacy of injected liquid silicone in the diabetic foot to reduce risk factors for ulceration: a randomized double-blind placebo-controlled trial. *Diabetes Care*. 2000;23:634–638.
16. Fountenteze JP, Simpson H, Kerrouche N. Hyaluronic acid filler injections under the metatarsal heads provide a significant and long-lasting improvement in metatarsalgia from wearing high-heeled shoes. *Dermatologic Surg*. 2018;44:994–1001.
17. Miller A. Filler in the foot: treatment of plantar fat pad atrophy with dermal fillers. *J Aesthet Nurs*. 2021;10:6–11.
18. Ali NS, Hoss E, Kollipara R, et al. Hyaluronic acid filler for treatment of metatarsal fat pad atrophy. *Dermatologic Surg*. 2021;47:1034–1035.
19. Salvatore L, Natali ML, Brunetti C, et al. An update on the clinical efficacy and safety of collagen injectables for aesthetic and regenerative medicine applications. *Polymers*. 2023;15:1020.
20. Billings E, Jr, May JW, Jr. Historical review and present status of free fat graft auto transplantation in plastic and reconstructive surgery. *Plast Reconstr Surg*. 1989;83:368–381.
21. Schoenhaus J, Gold J, McFarlane A. Pressure Offloading Injectable System (POIS): evaluating the use of an injectable dermal filler for offloading painful pressure points during weight bearing activities. *J Int Foot Ankle Foundation*. 2022;1:3–14.
22. Minter DM, Gusenoff BR, Gusenoff JA. Fat grafting for pedal fat pad atrophy in a 2-year, prospective, randomized, crossover, single-center clinical trial. *Plast Reconstr Surg*. 2018;142:862e–871e.
23. Gold JS. Early clinical experience with the use of a new allograft adipose matrix for foot fat pad restoration. *J Am Podiatr Med Assoc*. 2023;113:21–216.
24. Regulski MJ, Saunders MC, McCulloch SE, et al. Pilot study: human adipose tissue allograft for fat pad defects in patients with preulcerative lesions. *Plast Reconstr Surg Glob Open*. 2024;12:e6404.
25. Liposana® Package Insert. Britecye, Inc. 2025.
26. Rapisio E, Calderazzi F. Fat grafting for chronic heel pain following surgery for adult flatfoot deformity: pilot study. *Foot (Edinb)*. 2017;31:56–60.

27. Fontes T, Brandão I, Negrão R, et al. Autologous fat grafting: harvesting techniques. *Ann Med Surg (Lond)*. 2018;36:212–218.
28. Wederfoort JLM, Hebels SA, Heuts EM, et al. Donor site complications and satisfaction in autologous fat grafting for breast reconstruction: a systematic review. *J Plast Reconstr Aesthet Surg*. 2022;75:1316–1327.
29. Rohrich RJ, Sorokin ES, Brown SA. In search of improved fat transfer viability: a quantitative analysis of the role of centrifugation and harvest site. *Plast Reconstr Surg*. 2004;113:391–395; discussion 396.
30. Doornaert M, Colle J, De Maere E, et al. Autologous fat grafting: latest insights. *Ann Med Surg (Lond)*. 2018;37:47–53.
31. Kokai LE, Schilling BK, Chnari E, et al. Injectable allograft adipose matrix supports adipogenic tissue remodeling in the nude mouse and human. *Plast Reconstr Surg*. 2019;143:299e–309e.
32. Morag E, Cavanagh PR. Structural and functional predictors of regional peak pressures under the foot during walking. *J Biomech*. 1999;32:359–370.
33. Mens MA, Fassaert TF, Homan JH, et al. Sub-calcaneal planar fat pad assessment using dual-energy computed tomography: first experience in the diabetic foot. *Clin Biomech (Bristol)*. 2023;110:106126.
34. Maemichi T, Matsumoto M, Tsutsui T, et al. Functional morphologic changes of the heel fat pad and plantar fascia in patients with heel pain during weightbearing and nonweightbearing. *Foot Ankle Orthop*. 2024;9:24730114241247824.
35. Molines-Barroso RJ, García-Álvarez Y, García-Klepzig JL, et al. Differences in the sub-metatarsal fat pad atrophy symptoms between patients with metatarsal head resection and those without metatarsal head resection: a cross-sectional study. *J Clin Med*. 2020;9:794.
36. Ruane EJ, Minter DM, Wyse AJ, et al. Volumetric analysis in autologous fat grafting to the foot. *Plast Reconstr Surg*. 2019;144:463e–470e.